Table 1. Model performance after oversampling (mean held-out publication-grouped scores)

K	Model	Oversampling	Accuracy	Precision	Recall	F1-score	Valid folds
3	DT	ROS	0.7794	0.3758	0.3702	**0.3711**	6
3	NN	ROS	0.6567	0.3589	0.4060	**0.3414**	6
3	RF	ROS	0.8667	0.3691	0.4015	**0.3800**	6
3	SVM	ROS	0.8177	0.3059	0.3141	**0.3098**	6
3	DT	ADASYN	0.8219	0.3313	0.3760	**0.3449**	6
3	NN	ADASYN	0.6733	0.3665	0.4122	**0.3560**	6
3	RF	ADASYN	0.8930	0.3847	0.3766	**0.3746**	6
3	SVM	ADASYN	0.8082	0.2953	0.3006	**0.2977**	6
3	DT	SMOTE	0.8545	0.2993	0.3177	**0.3080**	6
3	NN	SMOTE	0.6787	0.3532	0.4044	**0.3451**	6
3	RF	SMOTE	0.8798	0.3416	0.3545	**0.3436**	6
3	SVM	SMOTE	0.8082	0.2953	0.3006	**0.2977**	6
4	DT	ROS	0.7212	0.2638	0.2791	**0.2605**	6
4	NN	ROS	0.6003	0.2382	0.2364	**0.2015**	6
4	RF	ROS	0.8446	0.2484	0.2650	**0.2563**	6
4	SVM	ROS	0.8374	0.2457	0.2610	**0.2530**	6
4	*DT*	*ADASYN*	*N/A*	*N/A*	*N/A*	***N/A***	*0*
4	*NN*	*ADASYN*	*N/A*	*N/A*	*N/A*	***N/A***	*0*
4	*RF*	*ADASYN*	*N/A*	*N/A*	*N/A*	***N/A***	*0*
4	*SVM*	*ADASYN*	*N/A*	*N/A*	*N/A*	***N/A***	*0*
4	*DT*	*SMOTE*	*N/A*	*N/A*	*N/A*	***N/A***	*0*
4	*NN*	*SMOTE*	*N/A*	*N/A*	*N/A*	***N/A***	*0*
4	*RF*	*SMOTE*	*N/A*	*N/A*	*N/A*	***N/A***	*0*
4	*SVM*	*SMOTE*	*N/A*	*N/A*	*N/A*	***N/A***	*0*
5	DT	ROS	0.6040	0.2869	0.3137	**0.2744**	6
5	NN	ROS	0.4838	0.2140	0.1772	**0.1769**	6
5	RF	ROS	0.7926	0.3653	0.3402	**0.3345**	6
5	SVM	ROS	0.7305	0.2919	0.2627	**0.2642**	6
5	*DT*	*ADASYN*	*N/A*	*N/A*	*N/A*	***N/A***	*0*
5	*NN*	*ADASYN*	*N/A*	*N/A*	*N/A*	***N/A***	*0*
5	*RF*	*ADASYN*	*N/A*	*N/A*	*N/A*	***N/A***	*0*
5	*SVM*	*ADASYN*	*N/A*	*N/A*	*N/A*	***N/A***	*0*
5	*DT*	*SMOTE*	*N/A*	*N/A*	*N/A*	***N/A***	*0*
5	*NN*	*SMOTE*	*N/A*	*N/A*	*N/A*	***N/A***	*0*
5	*RF*	*SMOTE*	*N/A*	*N/A*	*N/A*	***N/A***	*0*
5	*SVM*	*SMOTE*	*N/A*	*N/A*	*N/A*	***N/A***	*0*